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Research article

Exploring the disordered magnetic ground state and spin dynamics in the honeycomb oxide $Na_3Cu_2SbO_6$

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ABSTRACT

Copper-based oxide materials have emerged as promising candidates for exploring low-dimensional magnetism due to their unique electronic and magnetic properties. Among these, $Na_3Cu_2SbO_6$ stands out as a particularly intriguing compound for both experimental and theoretical investigations, owing to its distorted honeycomb lattice structure and the presence of spin-gap behavior. In this study, we conducted comprehensive magnetization, heat capacity, and muon spin relaxation (μSR) measurements on the layered honeycomb oxide $Na_3Cu_2SbO_6$ to elucidate its magnetic ground state and probe the complex spin dynamics in detail. $Na_3Cu_2SbO_6$ crystallizes in a C-centered monoclinic structure with the space group $C2/m$ (No. 12). Magnetization and specific heat measurements reveal the absence of long-range magnetic ordering down to 2 K. Furthermore, specific heat data exhibit a Schottky-like anomaly at low temperatures, indicative of short-range magnetic correlations, while no evidence of long-range ordering is observed. The temperature dependence of the muon spin relaxation rate λ suggests the presence of spin fluctuations or a quantum-disordered ground state in $Na_3Cu_2SbO_6$. The absence of oscillations in the zero-field (ZF) μSR spectra further confirms the lack of long-range magnetic order in the material. Longitudinal field (LF) μSR experiments provide additional insights, revealing that the spin fluctuations persisting at low temperatures are dynamic in nature. Our findings indicate that the disordered and fluctuating magnetic ground state in $Na_3Cu_2SbO_6$ can be attributed to magnetic frustration arising from competing ferromagnetic nearest-neighbor and antiferromagnetic next-nearest-neighbor interactions along the b axis within the ab plane. These results highlight the rich magnetic behavior of $Na_3Cu_2SbO_6$ and its potential as a model system for studying low-dimensional magnetism and quantum spin phenomena.

1. Introduction

Frustrated magnetism occurs in materials when spins interact via competing exchange interactions which cannot be fulfilled simultaneously resulting in ground state degeneracy [1]. Geometrical frustration mostly arises in triangular, kagome, hyperkagome and pyrochlore lattices [2,3]. Very strong frustration leads to spin liquid ground state with no long range magnetic order or static magnetism down to zero Kelvin [4]. The Kitaev honeycomb model is exactly solvable and predicts quantum spin liquid (QSL) ground state in honeycomb lattice [1–4]. Because of its unique combination of being experimentally relevant, exactly solvable and hosting a variety of different interesting gapped and gapless QSL phases, it is important to investigate honeycomb lattices compounds to find out new disorder free QSL. Some QSL candidates include organic anisotropic Mott insulators κ-$(ET)_2Cu_2(CN)_3$ [5] and $EtMe_3Sb[Pd(dmit)_2]_2$ [6], quantum kagome antiferromagnet $[ZnCu_3(OH)_6Cl_2]$ [7]. Other materials which falls under possible spin liquid candidates are Vesignieite $BaCu_3V_2O_8(OH)_2$ [8], Volborthite $Cu_3V_2O_7(OH)_2.2H_2O$ [9].

Alpichshev et al. pointed out the confinement-deconfinement transformation in α-Na_2IrO_3, which is an indicator of spin liquid behavior [10]. Hyperkagome lattice system $Na_4Ir_3O_8$ [11], harmonic honeycomb iridates α, β, and γ-Li_2IrO_3 [12], other iridates such as $H_3LiIr_2O_6$ [13] and $Ag_3LiIr_2O_6$ [14] are intriguing candidates for Kitaev physics due to Ir-d orbitals. The discovery of high-temperature Kitaev QSL in α-$RuCl_3$ has sparked a quest for non-Abelian QSL in multilayer oxides [15]. Li et al. observed atypical phonon effects in α-$RuCl_3$ via inelastic X-ray

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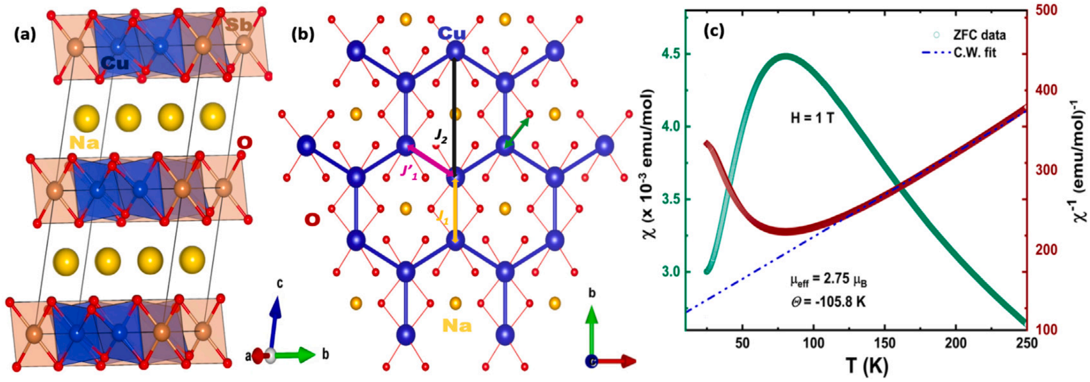

Fig. 1. (a) Honeycomb layers separated by Na atom layers. (b) Honeycomb geometry of $Na_3Cu_2SbO_6$ in the ab-plane (blue hexagon shows the underlying honeycomb structure) (c) Temperature evolution of susceptibility (green open circles) and inverse susceptibility (brown open diamonds). The blue dashed–dotted line shows a high-temperature Curie–Weiss fit. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

scattering, indicating Kitaev magnetic excitations [16]. Experimental studies in graphene report that massless Dirac fermions can originate from itinerant electrons, and a topological quantum spin liquid with exotic quasiparticles can be generated in spin-1/2 magnets, as predicted theoretically in the Kitaev model [17]. Kitaev provided a sophisticated theory of QSLs that resulted in a perfectly solvable spin-liquid model on the honeycomb lattice with significant spin–orbit interactions (SOI).

Honeycomb layered oxides are an evolving class of materials that exhibit interesting physical and electrochemical properties [18]. Because of the presence of mobile alkali atoms sandwiched between honeycomb slabs, these layered honeycomb oxides at high temperatures are endowed with fast ionic conduction as well as at room temperature [18,19]. Novel layered complex metal oxides with honeycomb structures such as $A_3M_2DO_6$ and $A_2M_2DO_6$ (A = Na, Li, K or coinage atoms like Ag and Cu; M = Mg, Ni, Fe, Co, Zn, Mn, Cu, and Cr; D denotes Ru, Te, Nb, Os, Sb, Ir, W, Bi, Ta) consists of alkali or coinage metal atoms [18]. These atoms are surrounded by blocks of transition metal oxide octahedra that are arranged like a honeycomb around pnictogen or chalcogen oxide octahedra [20]. These oxide materials find their applications in optics, electrochemistry, electromagnetism, quantum computation, superfast ionic conduction, topology, and materials science [21]. Vivanco et al. recently reported spin-flop type transition and signatures of Kitaev physics in an isostructural compound $Li_3Co_2SbO_6$ by neutron diffraction studies, and thermodynamic measurements [22]. Cu based low dimensional oxide materials are interesting for studying low-dimensional magnetism because they have $S = 1/2$ spin, which is guaranteed by the Cu $3d^9$ electron structure along with the underlying low dimensionality of magnetic model [23].

Miura et al. [24] have reported that the magnetic susceptibility of $Na_3Cu_2SbO_6$ exhibits a broad maximum at around 90 K which shows the existence of a potential spin gap state. Heat capacity measurements revealed a lack of long-range ordering down to 2 K. It crystallizes in C-centered monoclinic structure having space group $C2/m$ and it consists of Cu_2SbO_6 layers with Na layers between them. $Na_3Cu_2SbO_6$ has a honeycomb geometry (see Fig. 1a & b) of edge-sharing CuO_6 octahedra, which is immensely distorted due to the Cu^{2+} Jahn-Teller effect [25]. A combination of lattice distortion, an orbital arrangement, and frustration produces the quasi-one-dimensional structure in $Na_3Cu_2SbO_6$. In this compound, two dominant couplings of Cu^{2+} ions occur along the Cu–O–Cu (J_1) superexchange path and Cu–O–Sb–O–Cu (J_2) super-super exchange path (see Fig. 1b) [26].

We have recently investigated the spin dynamics of $Li_3Cu_2SbO_6$ by neutron diffraction, muon spin relaxation and inelastic neutron scattering experiments indicating the presence of a QSL like fluctuating ground state in $Li_3Cu_2SbO_6$ [27]. Motivated by the emergence of such exotic state in $Li_3Cu_2SbO_6$ we have performed zero-field (ZF) and longitudinal field (LF) muon spin relaxation (μSR) measurements in isostructural $Na_3Cu_2SbO_6$ to probe the static and dynamic spin fluctuations. Lack of oscillation in ZF asymmetry spectra and asymmetry value at$t = 0$ time is same for all spectra indicates the absence of any long range magnetic order. A platue like behavior in ZF relaxation rate indicates the presence of dynamic spin fluctuations also supported by LF-μSR data. These results reveal the intriguing magnetic behavior of $Na_3Cu_2SbO_6$, which, like $Li_3Cu_2SbO_6$, seems to host a fluctuating spin state. The lack of long-range magnetic order and the evidence for dynamic spin fluctuations imply that $Na_3Cu_2SbO_6$ may exhibit quantum spin liquid-like properties, warranting further theoretical and experimental investigation to fully understand its ground state.

2. Experimental details

A high quality polycrystalline $Na_3Cu_2SbO_6$ sample was prepared employing high temperature solid state reaction route. Na_2CO_3 (Alfa Aesar, 99.5%), Sb_2O_5 (Alfa Aesar, 99.998%), and CuO (Alfa Aesar, 97%) powder were taken as precursors and measured on bloating paper using highly sensitive weighing machine. The precursors are taken in a molecular ratio of 3:1:4 and ground for 4–5 h using an agate mortar-pestle for obtaining better stoichiometry. Acetone (Sigma Aldrich, 99%) was used in between grindings to get isotropic and homogeneous mixture. The complete reaction is given below;

$$3Na_2CO_3 + 4CuO + Sb_2O_5 \rightarrow 2Na_3Cu_2SbO_6 + 3CO_2 \uparrow$$

During sintering process large amount of Na_2CO_3 is sublimed due to its volatility, hence we added extra 10% Na_2CO_3 before the grinding was started. Putting 0.0014 GPa pressure for 5 min in hydraulic pellet press, the whole powder mixture was pelletized. Due to such pressure the contact surface area of the reactants are increased. Putting the highly dense pellets on alumina boat it was placed inside a box furnace and annealed (in air) at 850 °C for 5 h (ramping rate = 3 °C/min) followed by an intermittent grinding at 800 °C for 5 h and finally cooled down to room temperature at a rate of 2 °C/min.

Magnetization and heat capacity measurements were performed using a Quantum Design Physical Property Measurement System. A ^{3}He insert was utilized to measure heat capacity down to 100 mK. Zero-field (ZF) and longitudinal-field (LF) muon spin relaxation (μSR) experiments on $Na_3Cu_2SbO_6$ were carried out on the General Purpose Surface-Muon Instrument (GPS) spectrometer at the Swiss Muon Source (SμS) at the Paul Scherrer Institute (PSI), Switzerland. A powdered $Na_3Cu_2SbO_6$ sample was mounted on a high-purity copper plate (99.95%) with GE varnish [28] and cooled to 1.5 K using He-exchange gas in a standard He^4 cryostat. 100% spin-polarized positive muons were implanted into the sample, and the asymmetry of the resulting decay positrons was calculated using the formula:

$$A(t) = \frac{N_{\mathrm{forward}}(t) - aN_{\mathrm{backward}}(t)}{N_{\mathrm{forward}}(t) + aN_{\mathrm{backward}}(t)},$$

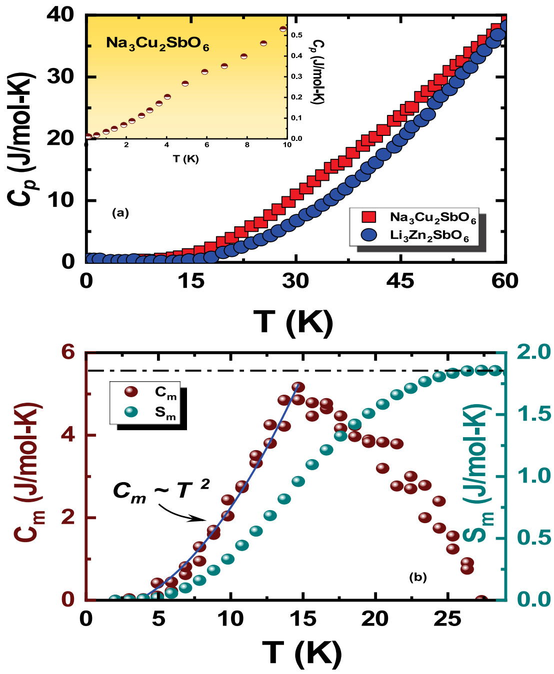

Fig. 2. (a) Temperature dependence of specific heat at constant pressure for $Na_3Cu_2SbO_6$ and isostructural non-magnetic analogue $Li_3Zn_2SbO_6$ (Taken from Ref. [27]). Within temperature range of 5–62 K, surplus specific heat is emitted demonstrating that the specific heat for $Na_3Cu_2SbO_6$ is greater than that caused by phonons. Inset shows the Schottky-type characteristics near 4 K. (b) Temperature dependent C_m (T) (brown balls) and entropy S_m (T) (dark cyan colored balls). The solid blue line is the outcome of approximation within the context of spin waves theory. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where a is the instrumental calibration constant, determined in the normal state with a nearly 2 mT transverse magnetic field [29]. $N_{\mathrm{forward}}(t)$ and $N_{\mathrm{backward}}(t)$ represent the number of positrons counted by the forward and backward detectors, respectively. All data collected were analyzed using Musrfit [30], a free, platform-independent framework for μSR data analysis.

3. Results and discussion

3.1. Magnetization

The temperature dependence of the magnetic susceptibility of $Na_3Cu_2SbO_6$ is shown in Fig. 1c. The green open circles and brown diamonds represent the zero-field cooled (ZFC) susceptibility (χ) and inverse susceptibility (χ^{-1}) data, respectively. The susceptibility was fitted using the model $\chi(T) = \chi_0 + \chi_{\mathrm{cw}}(T)$, where χ_0 is a temperature-independent term, and $\chi_{\mathrm{cw}}(T) = \frac{C}{T-\Theta}$ represents the Curie–Weiss term. The blue dashed–dotted line corresponds to the Curie–Weiss (CW) fit in the high-temperature regime. From the fit, we obtain a Curie constant $C = 0.95$ emu K/mol and a Weiss temperature $\Theta = -105.8$ K. In comparison, Derakhshan et al. reported a Curie constant $C = 1.019$ emu K/mol and a Weiss temperature $\Theta = -55$ K for $Na_3Cu_2SbO_6$ [31]. While our Curie constant aligns with their reported value, the difference in Weiss temperature may stem from the distinct temperature ranges used for fitting.

The effective magnetic moment, calculated using the formula $\mu_{\mathrm{eff}} = \frac{3k_BC}{N_A\mu_B^2}$, yields a value of $\mu_{\mathrm{eff}} = 2.75\,\mu_B$, which is consistent with previously reported values [31–33]. However, this value exceeds the spin-only moment of 1.73 μ_B typically observed for Cu^{2+} ions with $S = 1/2$. The higher observed magnetic moment could be attributed to additional factors, such as higher-order interactions or anisotropic effects within the system, though the exact mechanism remains unclear. For the isostructural lithium-based compound, a Curie–Weiss (CW) fit suggests that the number of effective Cu^{2+} spins per formula unit is 1.0 at high temperatures [27]. The negative Weiss temperature further indicates that antiferromagnetic (AFM) interactions are dominant at elevated temperatures.

The magnetic susceptibility shows a broad maximum around 80 K, suggesting the presence of a spin gap in $Na_3Cu_2SbO_6$, which is consistent with previous reports by Kuo et al. [34]. This feature is likely due to the 2D magnetism and short-range AFM character of the compound. Quantum Monte Carlo simulations by Schmitt et al. [23] also indicate that FM-AFM couplings are more dominant than AFM-AFM couplings in $Na_3Cu_2SbO_6$. A spin gap behavior has also been observed in another copper-based compound, $Na_2Cu_2TeO_6$, as demonstrated in susceptibility measurements by Miura et al. [24].

3.2. Heat capacity

Fig. 2(a) shows the temperature dependence of the specific heat of $Na_3Cu_2SbO_6$. To calculate the electronic heat capacity and magnetic entropy, we used the specific heat values of the non-magnetic isostructural counterpart $Li_3Zn_2SbO_6$ (Ref. [27]). The absence of a λ-type anomaly in the data rules out the presence of any long-range magnetic ordering down to 100 mK. The inset of Fig. 2(a) displays the temperature variation of C_p down to 100 mK, where a Schottky-like behavior is clearly observed around 5 K, suggesting the presence of short-range ordering in this system.

A similar behavior is seen in potential spin-liquid candidates, including $KCu_6AlBiO_4(SO_4)_5Cl$ [35], $ZnCu_3(OH)_6Cl_2$ [36], and $Li_3Cu_2SbO_6$ [27]. The magnetic entropy (S_m), associated with ordering, was calculated using the relation $S_m = \int_0^T \frac{C_p}{T}\,dT$ from the $C_p(T)$ data (see Fig. 2(b)). The light brown markers represent the magnetic specific heat, which exhibits a broad peak near 15 K. This feature could be related to the opening of a spin gap, possibly due to Cu–Cu dimerization, as observed in the isostructural $Li_3Cu_2SbO_6$ [27]. The dark cyan markers indicate the magnetic entropy, which reaches a maximum value of 1.85 J/mol-K around 28 K. The total entropy released, approximately 1.85 J/(mol K), is significantly lower than the 11.5 J/(mol K) predicted by mean-field theory for two Cu ions per formula unit, which would release $R\ln(2S+1)$ J/mol-K, where $S = 1/2$ for Cu.

We further analyzed the low-temperature behavior of $C_m(T)$ using the spin-wave (SW) model, assuming that the magnetic specific heat at low temperatures follows a $C_m \propto T^{d/n}$ power law for magnons [37], where d is the lattice dimensionality and n is the exponent in the dispersion relation $\omega \approx k^n$. Fitting the C_m data with this model yields values of $d = 2$ and $n = 1$ with high precision, suggesting that 2D magnons dominate at low temperatures. This supports the presence of significant magnetic frustration and short-range correlations in the $Na_3Cu_2SbO_6$ crystal. Inelastic neutron scattering and dispersion data further suggest that 1D spin chains of Cu^{2+} ions, with alternating short and long spacings along the b-axis, provide an explanation for the spin environment, as proposed by Miura et al. [38]. These findings underscore the inherent frustration arising from competing magnetic interactions along different bonds, leading to a complex and unconventional magnetic ground state in $Na_3Cu_2SbO_6$.

3.3. ZF and LF-μSR measurements

We conducted μSR experiments on $Na_3Cu_2SbO_6$ to probe the spin dynamics within its low-energy states. In these experiments, 100% spin-polarized muons are implanted into the material, where they undergo Larmor precession due to local magnetic fields at the implantation site. The muons decay with a characteristic lifetime of 2.2 μs, emitting positrons predominantly along the muon's instantaneous spin axis at the time of decay. Zero-field (ZF) or longitudinal-field (LF)-μSR is a powerful technique for investigating dynamic spin fluctuations or the

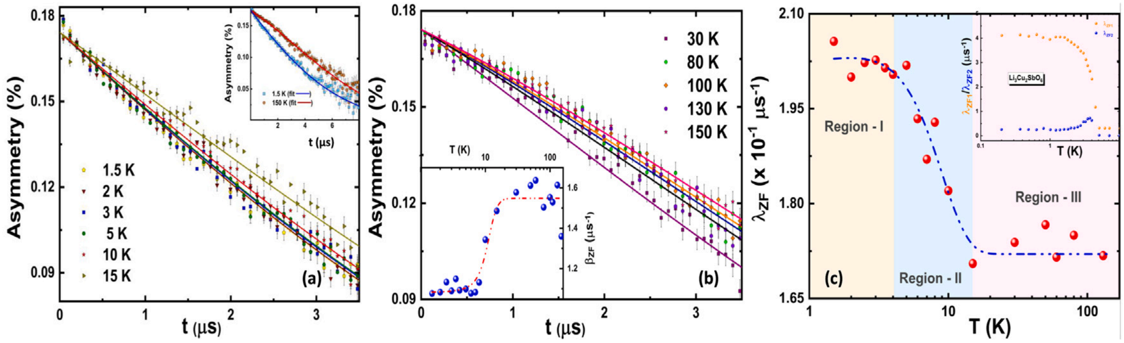

Fig. 3. Time dependent ZF-μSR asymmetry spectra at different temperatures in the temperature range (a) 1.5 K $\leq T \leq$ 15 K while inset represents asymmetry spectra at 1.5 K and 150 K upto a longer time scale of 8 μs without oscillation and (b) 30 K $\leq T \leq$ 150 K. Inset shows temperature variation of stretching exponent β. Red dashed line is guided to the eye. Curves fitted with the stretched exponent are the solid colored lines behind the data points. (c) Temperature dependence of relaxation rate λ, implies the existence of spin fluctuations in $Na_3Cu_2SbO_6$. Inset shows fast (λ_{ZF1}) and slow (λ_{ZF2}) relaxation rates of $Li_3Cu_2SbO_6$ [27] for comparison.

magnetic ground state of a material, and is frequently used to explore the behavior of quantum spin liquids (QSLs) or QSL-like materials. ZF spectra were recorded for powdered $Na_3Cu_2SbO_6$ in the temperature range of 1.5 K to 150 K.

Fig. 3(a) presents the ZF-μSR spectra in the temperature range of 1.5 K $\leq T \leq$ 15 K, revealing the local magnetic response of muons implanted at random stopping sites within the material. Fig. 3(b) shows the ZF-μSR asymmetry for temperatures between 30 K and 150 K. The inset of Fig. 3(a) displays the asymmetry data at both base temperature (1.5 K) and high temperature (150 K) for a longer time domain. The absence of oscillations in the ZF muon spectra suggests the absence of long-range magnetic order, consistent with observations in the isostructural compound $Li_3Cu_2SbO_6$ [39]. Notably, no fast relaxation components were observed, in contrast to what has been seen in isostructural compounds like $Li_3Cu_2SbO_6$ [40] and Li_4MgOsO_6 [41].

The ZF data were fitted using a stretched exponential function [27, 42–47]:

$$G_{ZF}(t) = C_0 \exp\left[\left(-\lambda_{ZF}t\right)^{\beta_{ZF}}\right] + C_b,$$

where λ_{ZF} is the muon relaxation rate, C_b represents the background asymmetry caused by muons in the silver sample holder, and C_0 denotes the initial asymmetry resulting from muons interacting with the sample. The background parameter C_b is temperature-independent. The exponent β_{ZF} accounts for the stretching of the relaxation, reflecting the cumulative effects of nuclear dipole fields in a disordered magnetic system. The stretched exponential function is commonly observed in a wide range of disordered magnetic systems [48].

In both Fig. 3(a) and (b), it is evident that the muon spin relaxation rate decreases with increasing temperature, as expected, since thermal energy dominates at higher temperatures, leading to reduced relaxation effects. Furthermore, at $t = 0$, all asymmetry curves converge to a single value in both figures, indicating the absence of long-range magnetic order in the system.

The inset in Fig. 3(b) reveals a gradual decline in β_{ZF}, suggesting a shift in spin dynamics. At higher temperatures, $\beta_{ZF} = 1.5$ indicates a mix of static and dynamic fluctuations, likely influenced by nuclear spins from elements like Na, Cu, and Sb, as well as rapidly fluctuating electronic spins from Cu^{2+}. As the temperature decreases, β_{ZF} transitions toward 1, signifying a dominance of dynamic spin fluctuations. Although such transitions are rare in Tomonaga–Luttinger spin-liquid systems, similar patterns have been observed in other candidates, such as $K_3Cu_3AlO_2(SO_4)_4$ [42]. In contrast, spin-glass systems typically exhibit a drop in β_{ZF} from 1 to 1/3 at the spin-freezing point [49]. In our case, the decrease in β_{ZF} at low temperatures is likely due to unpaired spins resulting from chain breakage in the disordered structure [50].

The temperature dependence of the relaxation rate is shown in Fig. 3(c), which is divided into three distinct regions. Region-I (pale yellow) corresponds to the temperature range 1.5 K $\leq T \leq$ 4 K, Region-II (light cyan) spans 4 K $\leq T \leq$ 15 K, and Region-III (pale pink) covers 15 K $\leq T \leq$ 150 K. A dashed–dotted blue line is included to guide the eye through the three regions.

From a visual inspection, it is clear that in Region-III (above 15 K), the relaxation rate becomes nearly temperature-independent, indicating the dominance of thermal fluctuations at high temperatures. In Region-II, as the temperature decreases, the relaxation rate increases sharply up to 4 K, suggesting the onset of spin fluctuations or a disordered magnetic state, where spins remain unfreezed. At higher temperatures, thermal fluctuations govern the system, but as the temperature decreases, quantum fluctuations become more prominent. This suggests that quantum effects play a crucial role in the formation of exotic states in $Na_3Cu_2SbO_6$.

This sharp rise in the relaxation rate stands in stark contrast to spin-glass systems, such as AgMn [51], CuMn [52], and Zn-doped $CuGeO_3$ [53], where spins freeze, and the asymmetry spectra exhibit a smooth, linear increase instead of the abrupt rise seen in $Na_3Cu_2SbO_6$.

As the temperature is further decreased, the relaxation rate reaches a plateau around 4 K and remains constant at lower temperatures, signifying a crossover to a plateau-like state in Region-I. This plateau reflects low-energy spin fluctuations. The subsequent rise in the relaxation rate in Region-II as the temperature decreases indicates a crossover to slower spin dynamics, likely due to the development of short-range spin correlations. The relaxation rate increases sharply as the temperature drops below 15 K, marking the crossover from the paramagnetic phase to a potential quantum spin-liquid (QSL) phase. This behavior is indicative of systems with QSL-like characteristics, where the spins do not order conventionally but instead continue to fluctuate even at low temperatures. Similar crossovers have been observed in other quantum spin-liquid candidates [54,55].

The persistent sluggishness of the spin dynamics as the temperature approaches zero is a characteristic feature often observed in quantum spin-liquid (QSL) ground states, especially in highly frustrated magnets. The plateau-like behavior of Cu^{2+} spins at 4 K supports the existence of short-range spin correlations and a dynamic ground state. A similar behavior is seen in the lithium-based isostructural compound, which exhibits analogous features down to 80 mK, as described using a stretched exponential function.

Other materials, such as quantum kagome AFM $ZnCu_3(OH)_6SO_4$ [54], $Nd_3Ga_5SiO_{14}$ [56], the $S = 1/2$ hyperkagome $Na_4Ir_3O_8$ [57], spin-liquid candidates like $Ca_{10}Cr_7O_{28}$ [58], and 2D triangular AFM $YbMgGaO_4$ [55], show similar relaxation rate behavior down to temperatures as low as 60 mK. This behavior further reinforces the presence of Cu^{2+} spin fluctuations in $Na_3Cu_2SbO_6$, which persist down to 1.5 K, in excellent agreement with previously reported NMR studies [59].

In the LF-μSR technique, the asymmetry in the spatial and temporal distribution of positrons emitted during muon decay is measured as a

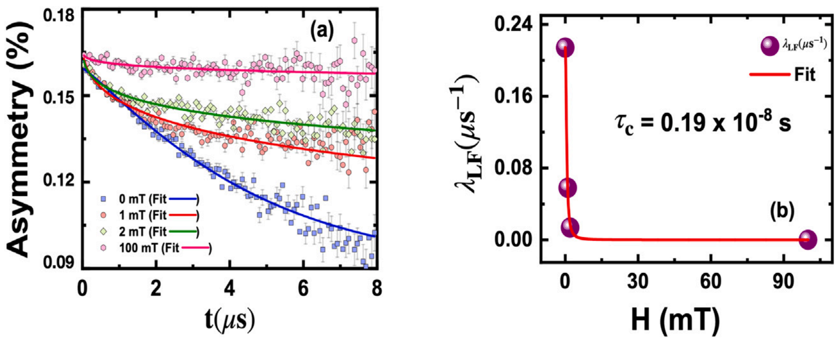

Fig. 4. (a) Time evolution of muon spin polarization of $Na_3Cu_2SbO_6$ with specified applied longitudinal fields at T = 1.5 K. Solid colored lines are fits to the asymmetry data. (b) Field dependence of LF-μSR spectra, solid red line denotes the fit using Redfield's equation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

function of time t after the muon implantation, while the longitudinal magnetic field $H \parallel \hat{\mathbf{z}}$ is varied. This method is particularly effective for probing magnetic behavior in materials with low magnetic moments and/or spatially disordered magnetism, which are common in geometrically and magnetically frustrated systems. To investigate the type of spin fluctuations in $Na_3Cu_2SbO_6$, we conducted LF analysis in an applied field range from 0 mT to 100 mT.

Fig. 4(a) displays the LF time spectra measured at fields of 0, 1, 2, and 100 mT. The asymmetry increases with increasing applied magnetic field, similar to observations in systems like AgMn [60] and $Yb_2Ti_2O_7$ [61]. However, the absence of saturation in the LF-μSR spectra at 100 mT indicates that the plateau-like behavior is not due to static magnetic fluctuations, which are typically seen in magnetically ordered systems. Therefore, this behavior is more likely attributed to dynamic fluctuations, as is often observed in proposed QSL-like materials.

We determined the local field H_{loc} and the fluctuation time τ_C by applying Redfield's equation [62] to fit the field dependence of the LF depolarization rate λ_{LF}, which is given by the expression:

$$\lambda_{\text{LF}} = \lambda_{LF_0} + \frac{2\gamma_\mu^2 H_{\text{loc}}^2 \tau_C}{1 + \gamma_\mu^2 H_{\text{ext}}^2 \tau_C^2}$$

Here, H_{ext} is the external longitudinal field, and H_{loc} represents the magnitude of the local field distribution, which governs the relaxation timeframe of the spin fluctuations τ_C.

Fig. 4(b) shows the variation of the relaxation rate with the longitudinal field. For $Na_3Cu_2SbO_6$, the fitting analysis yields an internal field distribution of $H_{\text{loc}} = 3.68\,\text{mT}$, using the muon gyromagnetic ratio $\gamma_\mu = 135.5 \times 2\pi\,\text{s}^{-1}\mu\text{T}^{-1}$. The corresponding correlation time is $\tau_C = 0.19 \times 10^{-8}$ s, and the longitudinal field relaxation rate is $\lambda_{\text{LF}}(0) = 0.004\,\mu\text{s}^{-1}$.

For the isostructural compound $Li_3Cu_2SbO_6$ [27], the fitting results show a smaller internal field distribution of $H_{\text{loc}} = 1\,\text{mT}$, a longer correlation time $\tau_C = 2.7 \times 10^{-8}$ s, and a slightly lower relaxation rate $\lambda_{\text{LF}}(0) = 0.003\,\mu\text{s}^{-1}$. These differences suggest that $Na_3Cu_2SbO_6$ has stronger local magnetic fields and faster spin dynamics compared to $Li_3Cu_2SbO_6$, which exhibits weaker local fields and slower dynamics.

3.4. Conclusion

In summary, we synthesized polycrystalline $Na_3Cu_2SbO_6$ using a high-temperature solid-state reaction method. The compound crystallizes in a C-centered monoclinic structure with the space group $C2/m$. Heat capacity measurements reveal the absence of long-range magnetic ordering down to 100 mK, while a Schottky-like anomaly near 5 K suggests the presence of short-range magnetic correlations. The magnetic entropy released accounts for approximately 16% (1.85 J/mol-K) of its saturation value, indicating significant magnetic frustration and short-range spin correlations. Zero-field muon spin relaxation (μSR) asymmetry spectra further support this, as all spectra collected in the temperature range of 1.5 K to 150 K converge to a single point at $t = 0$, confirming the existence of short-range ordering. The plateau-like behavior observed in the temperature-dependent relaxation rate $\lambda(T)$ may arise from competing exchange interactions between ferromagnetic (J_{FM}) and antiferromagnetic (J_{AFM}) couplings, originating from nearest-neighbor and next-nearest-neighbor Cu^{2+} ions, respectively. This competition could lead to a disordered or fluctuating magnetic state at low temperatures. To probe the nature of these fluctuations, longitudinal field (LF) μSR experiments were conducted. The LF-μSR spectra at 1.5 K demonstrate the presence of dynamic spin fluctuations rather than static ones, highlighting the complex magnetic behavior of $Na_3Cu_2SbO_6$. This study suggests that honeycomb-layered oxides with the general formula $A_3M_2DO_6$, where A = Na, K, Li, Ag, Cu; M = Cu, Fe, Co, Mg, Mn, Zn, Ni; and D = Sb, Ru, Bi, Ta, Te, Nb, Os, are promising candidates for exploring exotic magnetic states, such as quantum spin liquids or other fluctuating ground states. These materials, characterized by quasi-two-dimensional lattices with d^7 or d^9 electronic configurations, exhibit a rich variety of stacking orders and structural motifs. However, the spin dynamics and magnetic properties of many of these systems remain poorly understood, warranting further experimental and theoretical investigations to uncover their potential for hosting novel quantum phenomena.

CRediT authorship contribution statement

A.K. Jana: Writing – review & editing, Investigation, Formal analysis. **A. Bhattacharyya:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **D. Das:** Writing – review & editing, Visualization, Validation, Methodology, Investigation. **D.T. Adroja:** Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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